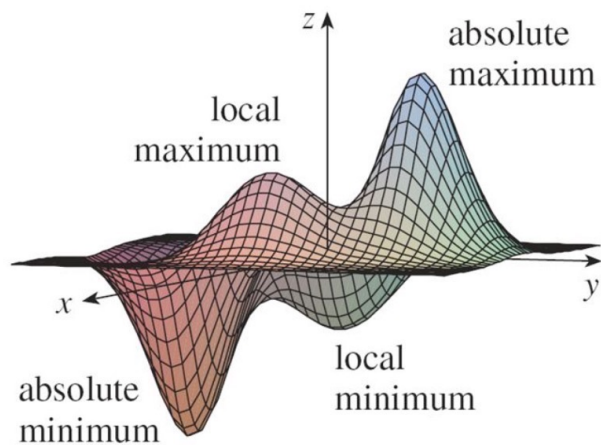




Review

Local extrema:



Thm: If F has a local maximum or minimum at (a, b) then $F_x(a, b) = 0$ and $F_y(a, b) = 0$.

Rmk: i.e. $\nabla F(a, b) = 0$

Def: If $\nabla F(a, b) = 0$, we call (a, b) a critical point of F .

Second derivatives test

Suppose (a, b) is a critical point of F (i.e. $F_x(a, b) = 0$ and $F_y(a, b) = 0$).

$$\text{Set } D = D(a, b) = F_{xx}(a, b)F_{yy}(a, b) - (F_{xy}(a, b))^2$$

- (a) If $D > 0$ and $F_{xx}(a, b) > 0$, then $F(a, b)$ - local min.;
- (b) If $D > 0$ and $F_{xx}(a, b) < 0$, then $F(a, b)$ - local max.;
- (c) If $D < 0$, then $F(a, b)$ is a saddle point of F .

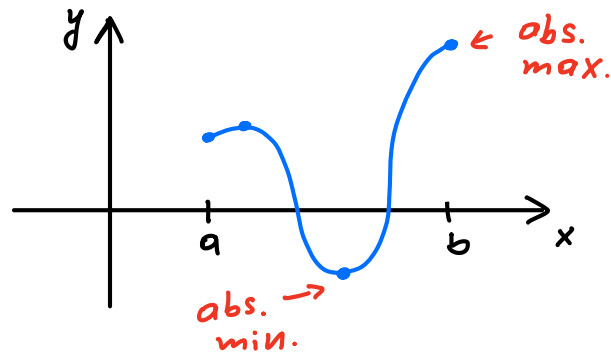
Rmk: If $D = 0$, the test gives no information.

Rmk:

$$D = \begin{vmatrix} F_{xx} & F_{xy} \\ F_{yx} & F_{yy} \end{vmatrix} = F_{xx}F_{yy} - (F_{xy})^2$$

Extrema on bounded regions

Recall: $f(x)$ continuous on $[a, b]$, has an absolute max. and an abs. min. on $[a, b]$.

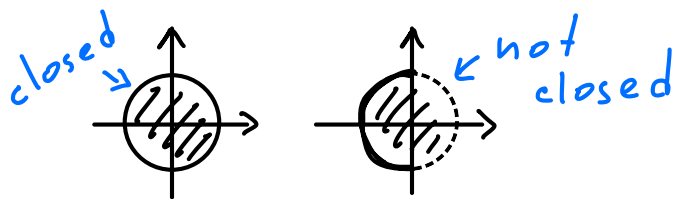


- We find them by evaluating f on critical points **AND** on endpoints.

$\mathcal{D}$ -closed set in $\mathbb{R}^2$

(i.e. contains boundary pts);

bounded set (contained in some disk)



To Find abs. max/min values of f on a closed, bounded $\mathcal{D}$:

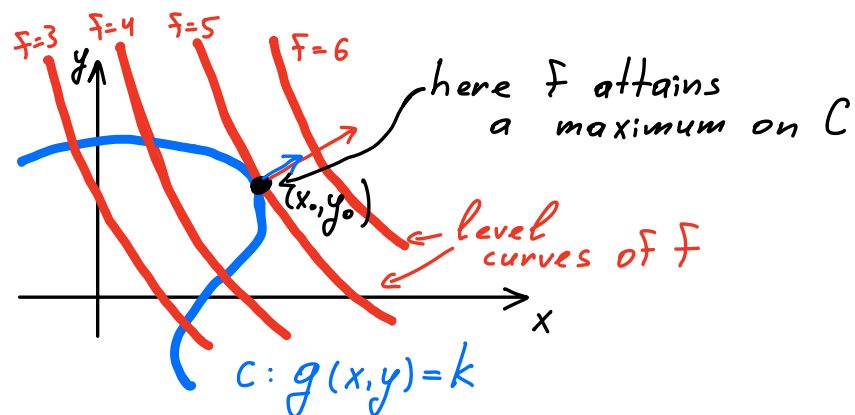
- ① Find values of f at critical points in $\mathcal{D}$;
- ② Find extreme values on the boundary of $\mathcal{D}$;
- ③ largest of ①, ② - abs. max.
smallest of ①, ② - abs. min.

f , continuous on a closed, bounded $\mathcal{D} \subset \mathbb{R}^2$,
attains an abs. max. value $f(x_1, y_1)$
and an abs. min. value $f(x_2, y_2)$ at some
points in $\mathcal{D}$.

Lagrange multipliers (one constrain)

Problem:

Find extreme values of $f(x,y)$
subject to constraint $g(x,y)=k$



level curves $g(x,y)=k$ and $f(x,y)=5$ touch at (x_0, y_0)

$$\Rightarrow \nabla f(x_0, y_0) = \underbrace{\lambda}_{\substack{\uparrow \text{a number, "Lagrange multiplier"}}} \nabla g(x_0, y_0)$$

$\uparrow$ a number, "Lagrange multiplier"

Method of Lagrange multipliers

To find maximum & minimum values of $f(x,y)$
subject to constrain $g(x,y)=k$:

① Find all values of x, y, λ such that

- $\nabla f(x,y) = \lambda \nabla g(x,y)$
- $g(x,y) = k$

$$\begin{cases} f_x = \lambda g_x \\ f_y = \lambda g_y \end{cases}$$

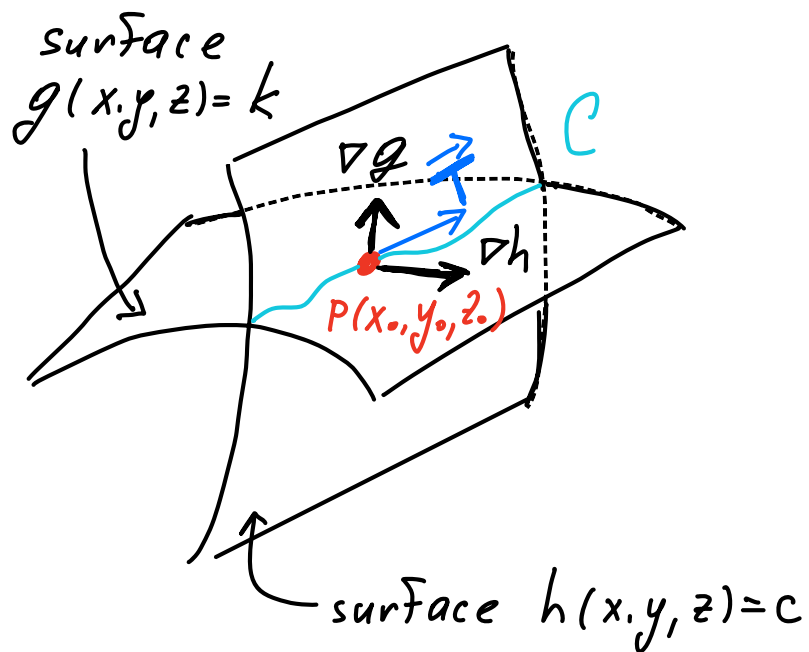
② evaluate f at all points found in (a)

maximum = largest value of f

minimum = smallest

Lagrange multipliers with two constraints

Problem: Find min/max values of $F(x, y, z)$ subject to two constraints: $\begin{cases} g(x, y, z) = k \\ h(x, y, z) = c \end{cases}$ $\swarrow \searrow$ define a curve C



If at $P(x_0, y_0, z_0)$ on C , F has a min or max, then directional derivative

$$D_{\vec{T}} F = 0, \quad (D_{\vec{T}} F = \nabla F \cdot \vec{T})$$

where $\vec{T}$ is the unit tangent vector along C at P .

So, $\nabla F(x_0, y_0, z_0)$ is orthogonal to $\vec{T}$

$\Rightarrow \nabla F|_P$ lies in the plane orthogonal to C at P .

This plane is defined by $\nabla g(x_0, y_0, z_0)$ and $\nabla h(x_0, y_0, z_0)$.

Therefore

$$\nabla F(x_0, y_0, z_0) = \lambda \nabla g(x_0, y_0, z_0) + \mu \nabla h(x_0, y_0, z_0)$$

$\nwarrow \nearrow$ Two Lagrange multipliers

Method of Lagrange multipliers

To find maximum & minimum values of $f(x, y, z)$
subject to constraints $g(x, y, z) = k$ and $h(x, y, z) = c$

① Find all values of x, y, z, λ, μ such that

- $\nabla f(x, y, z) = \lambda \nabla g(x, y, z) + \mu \nabla h(x, y, z)$
- $g(x, y, z) = k$
- $h(x, y, z) = c$

$$\begin{aligned} f_x &= \lambda g_x + \mu h_x \\ f_y &= \lambda g_y + \mu h_y \\ f_z &= \lambda g_z + \mu h_z \end{aligned}$$

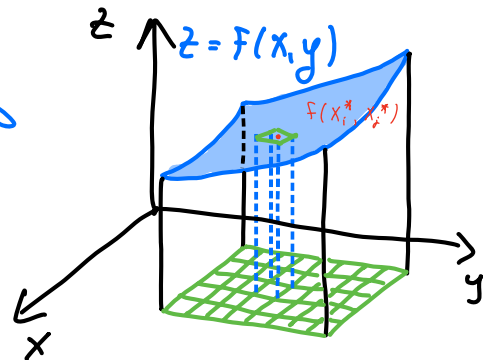
② evaluate f at all points found in (a)

maximum = largest value of f

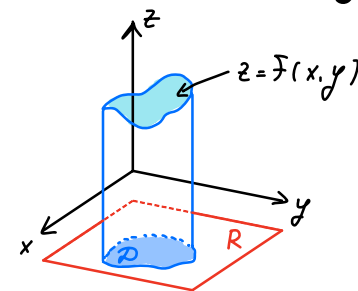
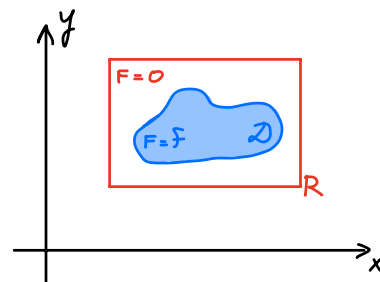
minimum = smallest

$\iint_R F(x,y) dA = \lim_{\substack{m \rightarrow \infty \\ n \rightarrow \infty}} \sum_{i,j} f(x_i^*, y_j^*) \underbrace{\Delta A}_{\Delta x \cdot \Delta y} \stackrel{\text{For } F \geq 0}{=} \text{volume of a solid that lies above the rectangle (general region) and below the surface } z = f(x,y).$

Double integrals:

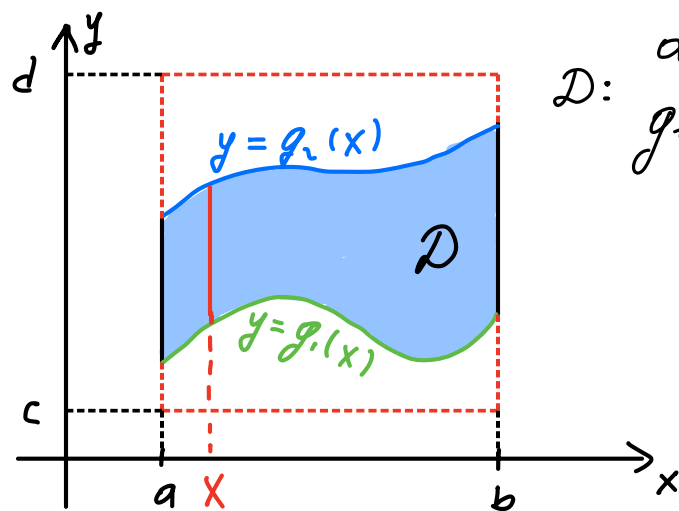


(x_i^*, y_j^*) - sample point in $[x_{i-1}, x_i] \times [y_{j-1}, y_j]$



Fubini's thm: if $f(x,y)$ continuous

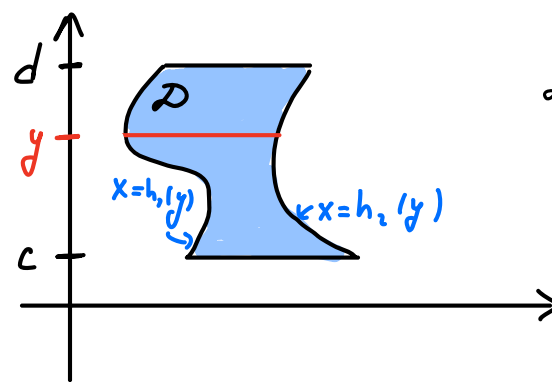
$$\iint_R f(x,y) dA = \int_a^b \int_c^d f(x,y) dy dx = \int_c^d \int_a^b f(x,y) dx dy$$



$D: a \leq x \leq b$
 $g_1(x) \leq y \leq g_2(x)$

$$\iint_D f(x,y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y) dy dx$$

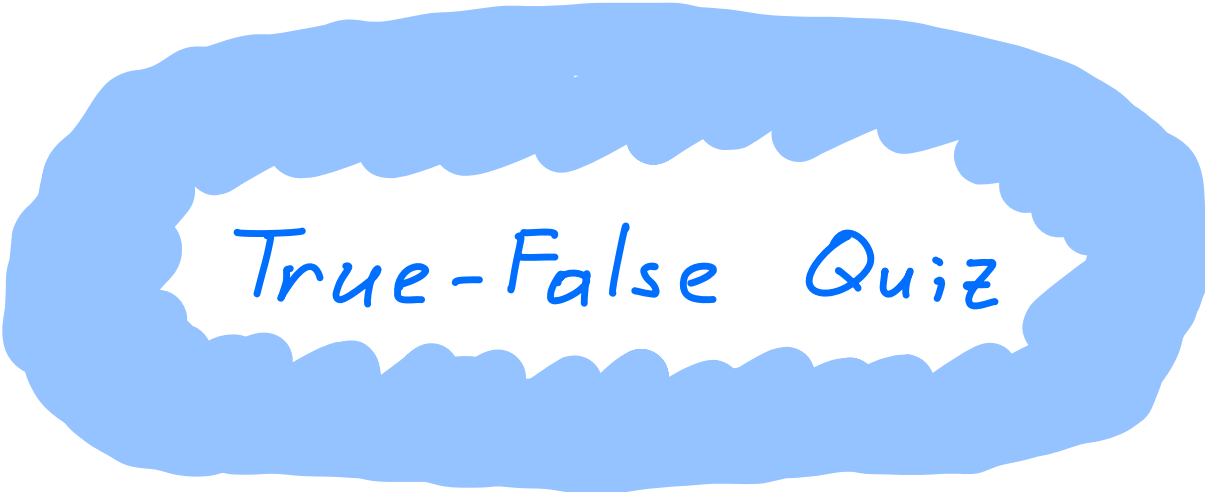
region of type I



$D: c \leq y \leq d$
 $h_1(y) \leq x \leq h_2(y)$

region of type II

$$\iint_D f(x,y) dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x,y) dx dy$$



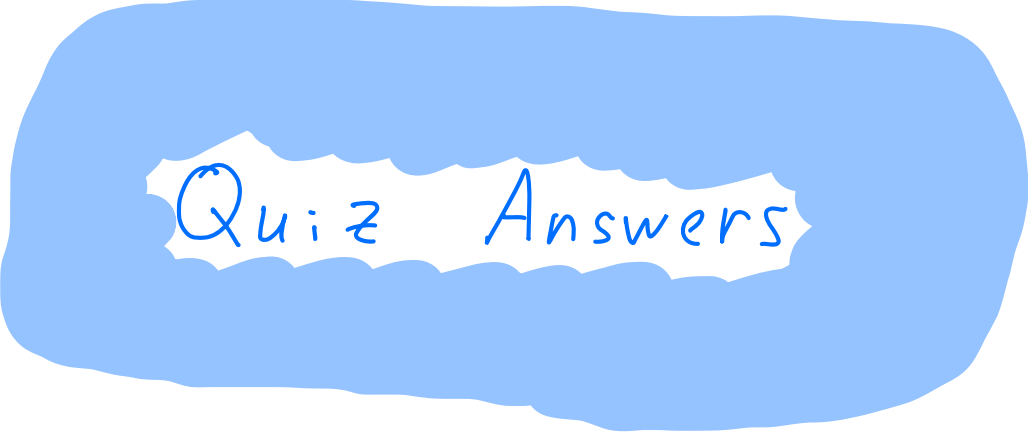
True-False Quiz

True-False Quiz:

- 1) IF a function has a local maximum at a point, then its gradient must be zero at that point.
- 2) To Find the absolute max/min of a continuous Function on a closed and bounded region, it is enough to check only the critical points inside the region.
- 3) The method of Lagrange multipliers can find both maxima and minima on a bounded region.
- 4) IF $\nabla f = \lambda \nabla g$ at a point, then the value at that point must be max or min of f under the constraint $g(x, y) = 0$.
- 5)
$$\int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx = \int_{g_1(x)}^{g_2(x)} \int_a^b f(x, y) dx dy$$
- 6) A double integral always gives the volume under the surface $z = f(x, y)$.

7) Let $\mathcal{D}$ be any region in the plane whose projection onto the x -axis is $[a, b]$.

$$\text{Then } \iint_{\mathcal{D}} (x^2 + e^x) dA = \int_a^b (x^2 + e^x) \cdot (\text{length of the vertical slice of } \mathcal{D} \text{ at } x) dx.$$



Quiz Answers

True-False Quiz: 1) IF a function has a local maximum at a point, then its gradient must be zero at that point. TRUE

2) To Find the absolute max/min of a continuous function on a closed and bounded region, it is enough to check only the critical points inside the region. FALSE

(must also check the boundary)

3) The method of Lagrange multipliers can find both maxima and minima ~~on a bounded region~~. FALSE

↳ subject to a constraint (which can be seen as a bdry of some region)

The method of Lagrange multipliers can be used to find max/min on the BOUNDARY of some region, but does NOT give information about the points inside the region.

4) IF $\nabla f = \lambda \nabla g$ at a point, then the value at that point must be max or min of f under the constraint $g(x, y) = 0$. FALSE

It also could be a saddle point.

5) $\int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y) dy dx \neq \int_{g_1(x)}^{g_2(x)} \int_a^b f(x,y) dx dy$

this is a number *this is some function $A(x)$*

FALSE

6) A double integral always gives the volume under the surface $z = f(x,y)$.

FALSE

This is true only when $f(x,y) \geq 0$.

7) Let D be any region in the plane whose projection onto the x -axis is $[a,b]$.

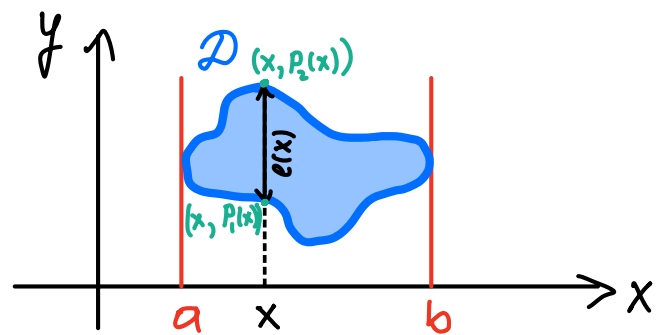
Then $\iint_D f(x) dA = \int_a^b f(x) \cdot (\text{length of the vertical slice of } D \text{ at } x) dx$.

Indeed, $\iint_D f(x) dA = \int_a^b \int_{p_1(x)}^{p_2(x)} f(x) dy dx = \int_a^b f(x) \ell(x) dx$

TRUE

$y f(x) \Big|_{p_1(x)}^{p_2(x)} = (p_2(x) - p_1(x)) f(x) = \ell(x) f(x)$

// since $f(x)$ does NOT depend on y //



Rmk: It is important that we integrate $f(x)$ here NOT $f(x,y)$.



Few more problems

1) Find the volume of the largest rectangular box in the 1st octant, with faces parallel to xy, yz, xz -planes and one vertex in the plane $x+2y+3z=6$.

2) Sketch the region $\mathcal{D}$ and rewrite $\iint_{\mathcal{D}} f(x,y) dA$ as $\int_{\boxed{?}}^{\boxed{?}} \int_{\boxed{?}}^{\boxed{?}} f(x,y) dx dy$ or $\int_{\boxed{?}}^{\boxed{?}} \int_{\boxed{?}}^{\boxed{?}} f(x,y) dy dx$,

where the region $\mathcal{D}$ is given by

a) $\mathcal{D} = \{ (x,y) \mid 1 \leq x \leq 2, 0 \leq y \leq \sqrt{3-x} \}$

b) $\mathcal{D} = \{ (x,y) \mid 0 \leq x \leq \pi, 0 \leq y \leq \sin x \}$

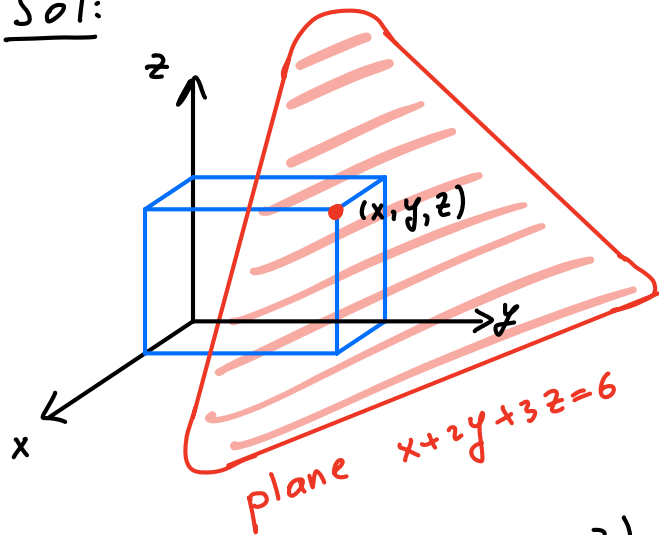
c) $\mathcal{D} = \{ (x,y) \mid 5y-14 \leq x \leq \frac{5}{4}y - \frac{11}{4}, 3 \leq y \leq 7 \}$

d) $\mathcal{D} = \{ (x,y) \mid 1 \leq x \leq 2, 2-x \leq y \leq \sqrt{1-(x-1)^2} \}$



(1) Find the volume of the largest rectangular box in the 1st octant, with faces parallel to xy, yz, xz -planes and one vertex in the plane $x+2y+3z=6$.

Sol:



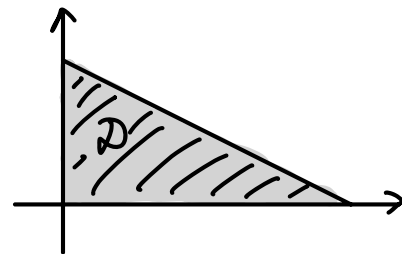
$$1) \quad z = \frac{6-x-2y}{3}$$

$$\text{Volume} = x \cdot y \cdot z$$

$$\text{Volume} = f(x, y) = xy \frac{6-x-2y}{3}$$

We want to maximize f in

$$D = \{ (x, y) : \begin{matrix} x+2y \leq 6 \\ x \geq 0, y \geq 0 \end{matrix} \}$$



$$2) \quad f_x(x, y) = y \frac{6-x-2y}{3} - \frac{1}{3}xy$$

$$f_y(x, y) = x \frac{6-x-2y}{3} - \frac{2}{3}xy$$

$$\begin{cases} y \frac{6-2x-2y}{3} = 0 \\ x \frac{6-x-4y}{3} = 0 \end{cases}$$

$x=0$ or $y=0 \leftarrow$ on the boundary of D

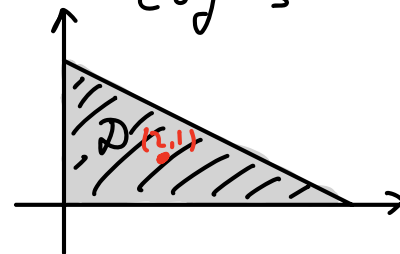
$$\begin{cases} 6-2x-2y=0 \\ 6-x-4y=0 \end{cases}$$

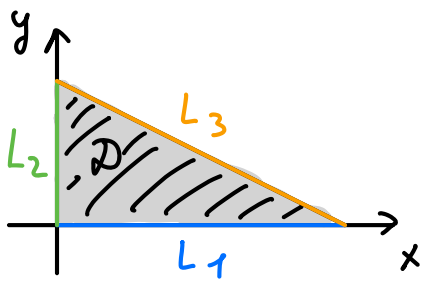
$$\Leftrightarrow \begin{cases} x+y=3 \\ x+4y=6 \end{cases}$$

$$\begin{cases} x=3-y \\ 3y=3 \end{cases}$$

$(2, 1)$ - critical point in the interior of D

$$f(2, 1) = 2 \cdot 1 \cdot \frac{6-2-2 \cdot 1}{3} = \frac{4}{3}$$





$$\text{On } L_1: f(x, 0) = 0, \quad 0 \leq x \leq 6$$

$$\text{On } L_2: f(0, y) = 0, \quad 0 \leq y \leq 3$$

$$\text{On } L_3: \begin{aligned} 0 &\leq y \leq 3 \\ x + 2y &= 6 \end{aligned} \quad x = 6 - 2y$$

$$f(x, y) = xy \frac{6 - x - 2y}{3} = (6 - 2y)y \frac{\overbrace{6 - (6 - 2y) - 2y}^{=0}}{3} = 0$$

$$\text{Abs. max: } f(2, 1) = \frac{4}{3}, \text{ box of size } \underset{x}{2} \times \underset{y}{1} \times \underset{z}{\frac{2}{3}}$$

(2) Sketch the region D and rewrite $\iint_D f(x, y) dA$ as

$\int \int_{\boxed{?} \boxed{?}} f(x, y) dx dy$ or $\int \int_{\boxed{?} \boxed{?}} f(x, y) dy dx$, where D is given by

a) $D = \{(x, y) \mid 1 \leq x \leq 2, 0 \leq y \leq \sqrt{3-x}\}$

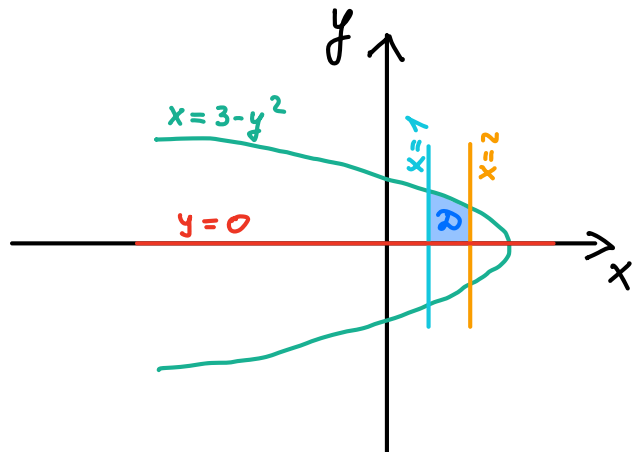
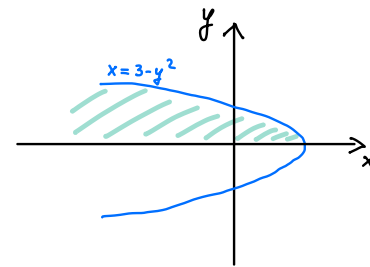
b) $D = \{(x, y) \mid 0 \leq x \leq \pi, 0 \leq y \leq \sin x\}$

c) $D = \{(x, y) \mid 1 \leq x \leq 6, 14+x \leq 5y \leq 11+4x\}$

d) $D = \{(x, y) \mid 1 \leq x \leq 2, 2-x \leq y \leq \sqrt{1-(x-1)^2}\}$

Sol: (2a) $0 \leq y \leq \sqrt{3-x} \Leftrightarrow$

$$\begin{cases} 0 \leq y \\ y^2 \leq 3-x \end{cases}$$

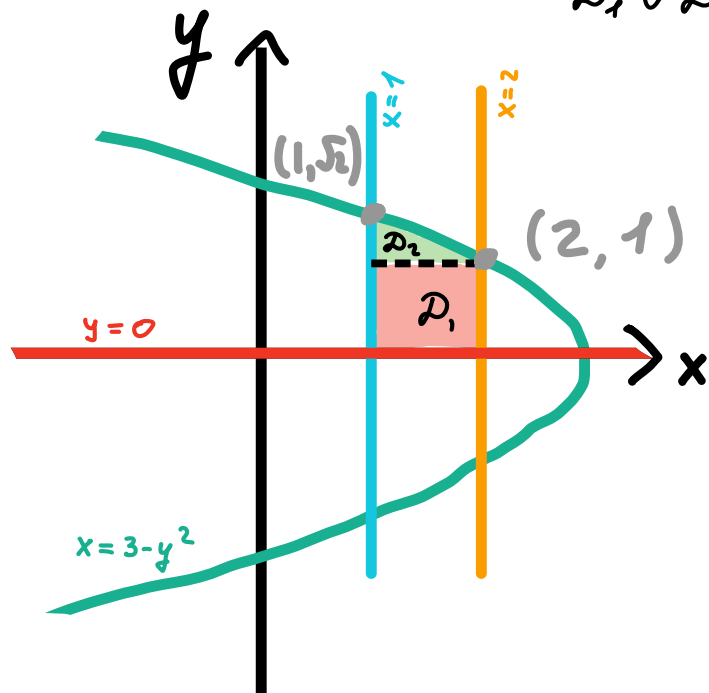


$$\iint_{\mathcal{D}} f(x,y) dA = \int_1^2 \int_0^{\sqrt{3-x}} f(x,y) dy dx \equiv$$

$$\int_0^1 \int_1^2 f(x,y) dx dy + \int_1^{\sqrt{2}} \int_1^{3-y^2} f(x,y) dx dy$$

$$= \iint_{\mathcal{D}_1} f(x,y) dA + \iint_{\mathcal{D}_2} f(x,y) dA$$

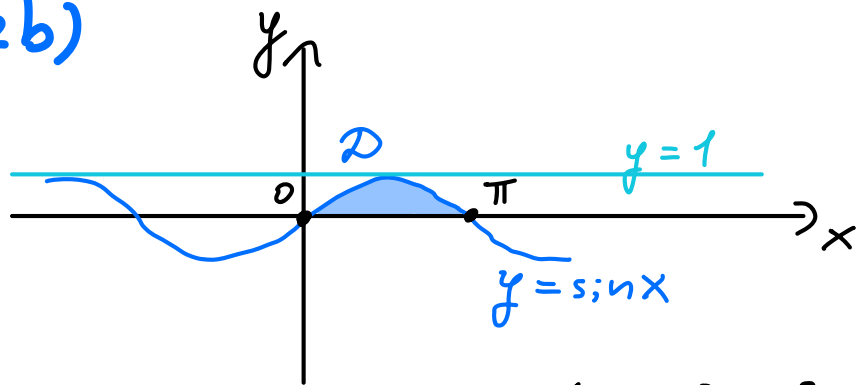
$$\mathcal{D}_1 \cup \mathcal{D}_2 = \mathcal{D}$$



$$\begin{cases} x=2 \\ x=3-y^2 \\ y \geq 0 \end{cases} \Rightarrow \begin{cases} x=2 \\ y=1 \end{cases}$$

$$\begin{cases} x=1 \\ x=3-y^2 \\ y \geq 0 \end{cases} \Rightarrow \begin{cases} x=1 \\ y=\sqrt{2} \end{cases} //$$

(2b)

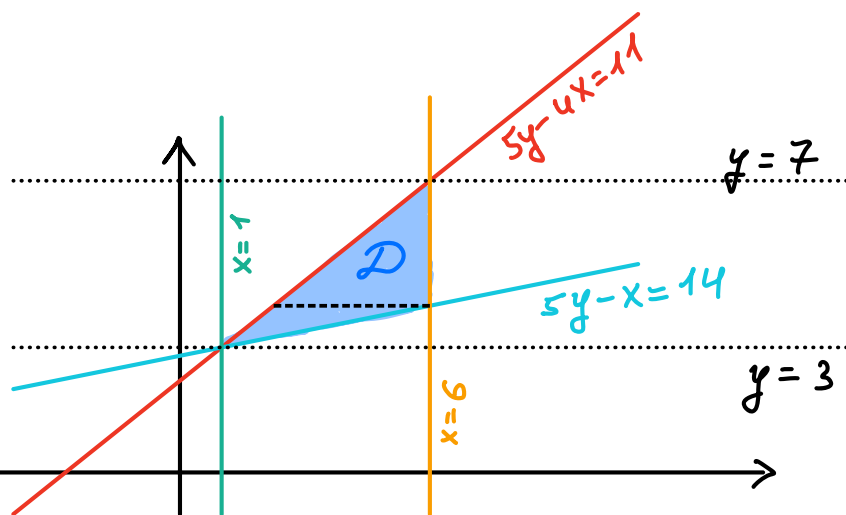


$$\iint_D F(x,y) dA = \int_0^\pi \int_0^{\sin x} F(x,y) dy dx$$

$$= \int_0^1 \int_{\arcsin(y)}^{\pi - \arcsin(y)} F(x,y) dx dy$$

// On $[0, \pi]$ $\sin x = y$ at $x = \arcsin(y)$ and $x = \pi - \arcsin(y)$ //

(2c)



$$\iint_D F(x,y) dA = \int_1^6 \int_{\frac{14+x}{5}}^{\frac{11+4x}{5}} F(x,y) dy dx$$

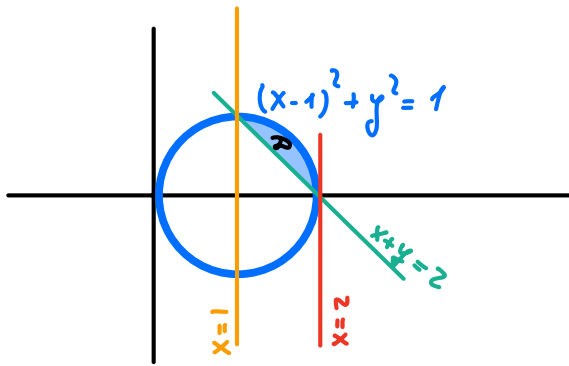
$$= \int_3^4 \int_{\frac{5y-11}{4}}^{5y-14} F(x,y) dx dy + \int_4^7 \int_{\frac{5y-11}{4}}^6 F(x,y) dx dy$$

$$(2d) \quad D = \{ (x, y) \mid 1 \leq x \leq 2, \ 2-x \leq y \leq \sqrt{1-(x-1)^2} \}$$

$$1 \leq x \leq 2 \Rightarrow 2-x \geq 0 \Rightarrow y \geq 0$$

$$\begin{cases} y \leq \sqrt{1-(x-1)^2} \\ y \geq 0 \end{cases} \Leftrightarrow \begin{cases} y^2 \leq 1-(x-1)^2 \\ y \geq 0 \end{cases} \Leftrightarrow \begin{cases} y^2 + (x-1)^2 \leq 1 \\ y \geq 0 \end{cases}$$

$$2-x \leq y \Rightarrow x+y \geq 2$$



$$\iint_D f(x, y) dA = \int_1^2 \int_{2-x}^{\sqrt{1-(x-1)^2}} f(x, y) dy dx$$

$$\begin{aligned} // \quad (x-1)^2 + y^2 &\leq 1 \\ \Rightarrow (x-1)^2 &\leq 1-y^2 \\ \Rightarrow x-1 &\leq \sqrt{1-y^2} // \end{aligned}$$

$$= \int_0^1 \int_{2-y}^{1+\sqrt{1-y^2}} f(x, y) dx dy$$